

Random Order:

1. In many electron atoms, assuming L-S coupling holds, list all the possible spectral (^{2S+1}L) and fine structure terms ($^{2S+1}L_J$) terms for the electronic configurations: $ns\ n's$, $ns\ n'p$, nd^2 , and $np\ n'p\ n''p$.
2. In many electron atoms, assuming j-j coupling holds, list all the possible terms (j_1, j_2) $_J$ for the electronic configurations: $ns\ n'p$, $(nl\ 3/2)^2$.
3. The Runge-Lenz operator $\vec{Q}$ given by

$$\vec{Q} = \frac{4\pi\epsilon_0}{2\mu Z_e^2} (\vec{L} \times \vec{p} - \vec{p} \times \vec{L}) + \frac{\vec{r}}{r}$$

Show that $[H, \vec{Q}] = 0$ and $\vec{L} \cdot \vec{Q} = 0$, where H is the hydrogen Hamiltonian ($\hat{p}^2/2\mu - Ze^2/(4\pi\epsilon_0 r)$).

4. Show that $M_{ab} = -M_{ba}^*$ where ($\hat{e}$ is polarisation unit vector)

$$M_{ba} = \langle \psi_b | e^{i\vec{k} \cdot \vec{r}} \hat{e} \cdot \vec{\nabla} | \psi_a \rangle$$

5. Write the velocity and acceleration forms of the dipole matrix elements.
6. Consider an electron in the electrostatic field of a nucleus of charge Ze and of mass number A . Assuming nuclear charge is distributed uniformly within a sphere of radius $R = r_0 A^{1/3}$ where $r_0 \approx 1.2 \times 10^{-15} m$
 - (a) Show that

$$\begin{aligned} V(r) &= \frac{Ze^2}{8\pi\epsilon_0 R} \left(\frac{r^2}{R^2} - 3 \right) & r \leq R \\ &= -\frac{Ze^2}{4\pi\epsilon_0 R}; & r \geq R \end{aligned} \quad (1)$$

- (b) Assuming $H' = V(r) - V_H(r)$, where $V_H(r)$ is hydrogenic potential energy with nuclear charge Z , estimate the first order energy shift for $r \leq R$.
7. Draw the normal Zeeman splitting energy levels for $n'd \rightarrow np$ with $m_s = 1/2$ and marks the possible multiplets (assuming electric dipole transitions).
 8. Show that the perturbative correction $\xi(r) \vec{L} \cdot \vec{S}$ does not connect the degenerate states with $m_l = +1$, $m_s = -1/2$ and $m_l = -1$, $m_s = +1/2$
 9. Problem 6.6 (B&J, 2nd edition) – Hamiltonian of a free electron in uniform magnetic field $\vec{B} = B_0 \hat{z}$.

10. Problem 7.1 (B&J, 2nd edition) – Eigenfunction $\psi(r_1, r_2)$ as symmetric and anti-symmetric...
11. Problem 7.2 (B&J, 2nd edition) – action of the $\hat{S}^2$ operator ...
12. Problem 7.3 (B&J, 2nd edition) – average values if $r_1^2 + r_2^2$...
13. Problem 7.4 (B&J, 2nd edition) – J_{nl} & K_{nl} for $n=2, l=0, 1$...
14. Show that $\hat{A}^2 = \hat{A}$ where

$$\hat{A} = \frac{1}{N!} \sum_P (-1)^P P$$

15. In Hartee-Fock treatment, show that for a given potential, the two spin orbitals u_λ and u_ν such that $E_\lambda \neq E_\nu$ are orthogonal i.e. $\langle u_\lambda | u_\nu \rangle = 0$.